# MOHAMED MUFASSIR


+94 77 423 3426 | mufassirriswan@gmail.com| Thihariya, Sri Lanka
mufassir.dev | linkedin.com/in/mohamedmufassir | github.com/M-Mufassir


## PROFESSIONAL SUMMARY


3rd-year Computer Systems Engineering undergraduate (SLIIT) specializing in embedded systems, robotics, and real-time control systems. Proven ability to design and build intelligent hardware-software systems — from AVR-programmed autonomous robots to IoT-connected vehicles with live video streaming. Supplemented by industry internship experience in software development. Passionate about robotics, industrial electronics, and building embedded solutions that operate reliably in the real world.


## EDUCATION

**BSc (Hons) in Computer Systems Engineering** Nov 2023 – Expected Nov 2027

Sri Lanka Institute of Information Technology (SLIIT) | GPA: Upper Second Class (3.0–3.5)

**Relevant Modules:** Real-Time Systems, Digital Logic Design, Control Theory, Computer Architecture, Data Structures & Algorithms, Computer Networks

**Advanced Level Examination** – Physical Science Stream 2022 (2023)

Combined Mathematics – B , Physics – C, Chemistry - C

## EXPERIENCE

**Software Engineering Intern — ZILLIT (Startup), Nittambuwa** *Nov 2025 – Feb 2026*

- Led sole development of a cross-platform Point of Sale (POS) desktop application using Electron.js and TypeScript, designed as part of a broader IoT ecosystem (desktop + mobile integration)
- Built the front-end of an enterprise-level financial management system using Vue.js, integrating with a Java Spring Boot backend
- Worked with React, TypeScript, Vue.js, Electron.js, PostgreSQL, and Docker in a fast-paced startup environment
- Collaborated on system architecture decisions for scalable, production-ready software

**Freelance Web Developer — Self-Employed, Local & Online** *2024 – Present*

- Designed and developed business websites for local clients including a jewelry shop, hardware store, restaurant, and an indoor playground booking platform
- Delivered end-to-end solutions covering UI design, branding, and front-end development
- Technologies used: React, HTML/CSS, JavaScript

## PROJECTS

**PID-Based DC Motor Speed Control System (Raspberry Pi + Python)** *2026*

- Designed a real-time closed-loop control system using PID algorithm with encoder feedback for precise RPM measurement and regulation
- Extended the project with a Python desktop GUI application to visualize live motor behavior, RPM graphs, and PID tuning parameters in real time
- Achieved stable, accurate motor control — successfully demonstrated and documented system performance
- Stack: Python, Raspberry Pi, PID control, encoder interfacing, data visualization

**AVR Autonomous Line-Following Car with Smart Parking (Arduino Uno)** *2026*

- Programmed an autonomous line-following robot using AVR (AT mega) in Microchip Studio with 3 IR sensors for line tracking and crossline detection logic
- Implemented a parking bay detection system using an IR receiver — upon single crossline detection, the car enters parking search mode, detects the bay, and parks autonomously
- On double crossline detection, the car exits parking mode and resumes normal line following, demonstrating a full FSM-based behavioral control system
- Stack: AVR (AT mega), Arduino Uno, Microchip Studio, Embedded C, IR sensors

**ESP32-CAM Remote-Controlled Car with Live Video Streaming** *2025*

- Built a WiFi-controlled RC car with live video streaming using the ESP32-CAM module as both the access point and video server
- Developed a mobile app and a web-based controller (live on Vercel) allowing real-time driving and camera feed viewing from any device on the network
- Integrated live MJPEG video stream with directional controls, synchronizing car movement and video feed over ESP32's internal WiFi
- Stack: ESP32-CAM, Embedded C, JavaScript, Vercel, WiFi/HTTP streaming

**IoT Smart Surveillance & Access Control System (ESP32)** *2024*

- Built a smart surveillance system integrating camera module with Telegram bot for real-time intrusion detection and remote door lock/unlock via commands
- Implemented UART/I2C communication protocols and real-time IoT messaging for reliable remote control
- Stack: ESP32, C/Embedded C, Telegram Bot API, IoT communication

**FSM-Based Sequence Detector** *2024*

- Designed a pseudo-random sequence generator using XOR feedback logic and PISO shift register serialization
- Implemented a Finite State Machine (FSM) for pattern detection with synchronized clocking

**Learning Management System (Spring Boot)** *2024*

- Developed RESTful backend APIs for authentication and course management with scalable database integration

## TECHNICAL SKILLS

**Languages:** C/C++, Embedded C, AVR Assembly, Python, JavaScript, TypeScript

**Frontend:** React, Vue.js, Electron.js, HTML/CSS

**Backend & DB:** Spring Boot (Java), Node.js, PostgreSQL, Docker (basic)

**Embedded & Hardware:** AVR (AT mega), Arduino, ESP32, Raspberry Pi, UART / SPI / I2C protocols, Linux OS, Oscilloscope & lab instruments, Digital Logic Design, Sensors & Actuators, PCB Design, 3D CAD Design

**Tools:** Git, Arduino IDE, Proteus, MATLAB, Fuson 360, VS Code

## CERTIFICATIONS & LEARNING

- **Introduction to Embedded Systems Software and Development Environments** – University of Colorado Boulder (Coursera) - embedded software development workflows, GCC toolchain, Linux development environments, version control using Git, and ARM-based embedded system concepts

- **The Complete Full-Stack Web Developer Bootcamp** — Dr. Angela Yu, Udemy (HTML, CSS, JS, Node.js, React, PostgreSQL)

## INTEREST AREAS

Embedded Systems | IoT | Robotics & Control Systems | Network Engineering | Telecommunications | Industrial Electronics | Software Engineering

## REFERENCES

**Prof. Anuradha Jayakody**

Head, Industry Engagement Unit , Department of Computer Systems Engineering – SLIIT, Malabe, Sri Lanka

Phone: +94 714900228 , Email: anuradha.j@sliit.lk

**Mr. Dinith Primal**

Head, Department of Computer Systems Engineering – SLIIT, Malabe, Sri Lanka

Phone: +94 766757270, Email: dinith.p@sliit.lk